

Designing an Emotionally Adaptive AI Companion Using Voice Emotion Recognition and Long-Term Emotional Memory

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1. Introduction

AI assistants such as Siri, Google Assistant, ChatGPT, and Replika are widely used for daily communication and emotional interaction. However, existing systems detect emotion mainly from text, lack persistent emotional memory across sessions, and provide generic, non-personalized responses. This research proposes an emotionally adaptive AI companion that combines voice-based emotion recognition with a structured long-term emotional memory and personalized emotional profiling, enabling more accurate, explainable, and individually tailored emotional support than current AI companion systems.

1.1 Summary of Literature

Recent studies confirm that AI companions such as Replika and ChatGPT are increasingly used for emotional support and coregulation [1], and that AI companions and AI assistants are converging into overlapping use cases [2]. However, three specific limitations persist. First, emotion detection in these systems is predominantly text-based; voice paralinguistic cues such as pitch, tone, and speech rate are largely unused. Second, memory is either session-limited (ChatGPT) or stores raw conversation logs without structured emotional

context (Replika), preventing identification of emotional patterns over time. Third, no system provides an explainable or quantifiable emotional profile users receive emotionally toned replies, but never an interpretable record of their own emotional trends. These three gaps define the scope of this research.

1.2 Problem Definition

Existing AI companion systems lack the ability to recognize emotion from vocal cues, retain emotional context across sessions in a structured, queryable form, and explain how a user's emotional state evolves over time. Current systems cannot answer questions such as "has the user become more anxious this month?" because emotional data is not stored as a structured, analyzable timeline. This research addresses this gap by building a voice emotion recognition pipeline coupled with a structured emotional memory architecture supporting trend analysis and personalized response generation, a combination not present in Replika, ChatGPT, or comparable systems.

1.3 Research Questions

RQ1: Does voice-based emotion recognition achieve significantly higher accuracy in identifying user emotional states than text-only emotion detection?

RQ2: Does structured long-term emotional memory with personalized profiling significantly improve perceived personalization, trust, and companionship compared to a memoryless baseline?

RQ3: Does an emotionally adaptive AI companion integrating both components achieve higher overall user satisfaction than a conventional AI assistant?

1.4 Hypotheses

H1: Voice-based emotion recognition achieves significantly higher accuracy in identifying user emotional states than text-only emotion detection.

H2: A structured long-term emotional memory with personalized emotional profiling significantly improves perceived personalization, trust, and companionship compared to a memoryless baseline.

H3: An emotionally adaptive AI companion integrating voice emotion recognition and structured emotional memory achieves significantly higher overall user satisfaction than a conventional (text-only, memoryless) AI assistant.

2. Goal of the Project

The goal is to develop an emotionally adaptive AI companion that recognizes user emotion from voice, maintains a structured long-term emotional memory, and generates personalized, explainable responses based on a user's evolving emotional profile. The scope is narrowed to three measurable components: voice emotion recognition, structured emotional memory with trend analysis, and personalized response generation. Clinical mental health diagnosis, crisis intervention, and professional therapeutic

features are explicitly outside the scope of this research.

3. Aims and objectives

Aim: To design and evaluate an emotionally adaptive AI companion that combines voice emotion recognition with explainable, longitudinal emotional memory to deliver measurably more personalized support than existing AI companions.

- **Objective 1:** To develop and evaluate a voice emotion recognition model (CNN-LSTM and Wav2Vec2-based) and benchmark it against text-only emotion detection to test H1.
- **Objective 2:** To design and implement a structured long-term emotional memory and personalized profiling architecture, and evaluate its effect on personalization, trust, and companionship to test H2.
- **Objective 3:** To integrate both components into a complete AI companion system and evaluate it against a conventional AI assistant baseline through a controlled user study to test H3.

4.Expected Outcomes

This project will deliver five measurable outcomes aligned with the stated hypotheses: (1) a trained and validated voice emotion recognition engine, benchmarked against a text-only baseline to test H1; (2) a structured long-term emotional memory framework with personalized emotional profiling, evaluated for its effect on personalization, trust, and companionship to test H2; (3) a fully integrated emotionally adaptive AI companion system; (4) a controlled user study comparing the integrated system against a conventional AI assistant baseline to test H3; and (5) a statistical analysis report confirming or rejecting H1, H2, and H3 using the metrics defined in Section 5.5. Success

is defined as achieving statistically significant results supporting the three hypotheses.

5. Proposed methodology

5.1 Voice Emotion Recognition (tests H1).

Raw audio will be preprocessed using Librosa to extract MFCCs, pitch, and energy features. A CNN-LSTM model will be trained to classify speech into six emotion categories (happy, sad, angry, fear, neutral, surprise), with a Wav2Vec2 transformer fine-tuned as a comparison architecture. To test H1, the voice-based model's classification accuracy will be directly benchmarked against a text-only emotion classifier (DistilRoBERTa on transcribed text) on the same evaluation set.

5.2 Long-Term Emotional Memory and Profiling (tests H2).

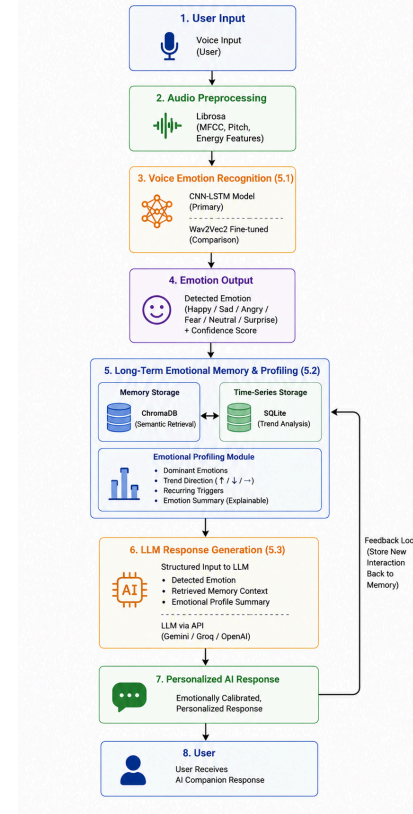
Each interaction is stored as a structured record timestamp, transcribed text, detected emotion with confidence score, and a semantic embedding (sentence-transformers) using a dual storage architecture combining ChromaDB (semantic retrieval) and SQLite (time-series trend analysis). A personalized emotional profiling module aggregates this history into an interpretable summary of dominant emotions, trend direction, and recurring triggers. To test H2, a memory-enabled condition will be compared against a memory-disabled (session-only) condition, measuring personalization, trust, and companionship via validated survey scales.

5.3 Integrated System and Response Generation (tests H3).

The detected emotion (5.1), retrieved memory context, and emotional profile (5.2) are passed as structured input to an LLM via API, generating a personalized, emotionally calibrated response. To test H3, the fully integrated system will be compared against a conventional AI assistant baseline (a memoryless, text-only chatbot using the same

underlying LLM) in a within-subject user study, isolating the effect of the proposed architecture from the base LLM itself.

5.4 System Architecture.



5.5 Evaluation Metrics. H1 will be tested using Accuracy, Precision, Recall, and F1-score, comparing voice-based and text-only emotion classifiers. H2 will be tested using validated personalization, trust, and companionship survey scales (5-point Likert), compared between memory-enabled and memory-disabled conditions. H3 will be tested using overall user satisfaction scores compared between the integrated system and the conventional baseline. All hypotheses will be evaluated using a paired t-test or Wilcoxon signed-rank test (depending on normality) at $\alpha = 0.05$, with a minimum sample size of $n=20$ participants.

6. Resource requirements

Hardware: Personal computers with internet access; GPU access via Google Colab/Kaggle for model training. **Software:** Python, TensorFlow/PyTorch, Scikit-learn, Librosa, HuggingFace Transformers (Wav2Vec2), VS Code. **Datasets:** RAVDESS and TESS speech-emotion datasets. **Cloud:** Google Colab and Kaggle for GPU-based training. **Other Tools:** LLM API (Gemini/Groq) for response generation; ChromaDB for vector memory; GitHub for version control.

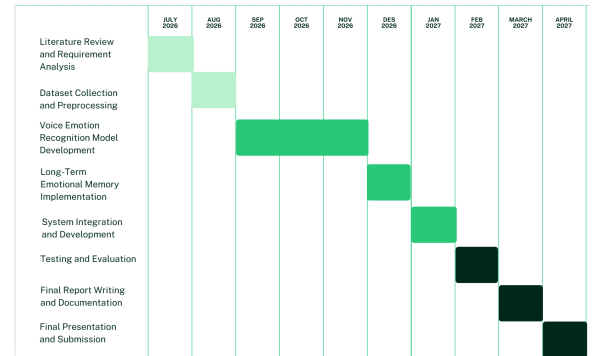
7. Ethics and AI Compliance

This project involves voice recordings and emotional data, requiring strict ethical handling. Informed consent will be obtained from all user study participants before data collection, with a clear explanation of how voice and emotional data will be used. All collected data will be anonymized, encrypted at rest, and used solely for research purposes. A defined data retention policy will limit storage duration, with participants able to request data deletion at any time. The system is explicitly designed as an emotional support tool and will not be presented as a substitute for professional mental health diagnosis or treatment. Ethics approval will be obtained from the SLTC Ethics Review Committee prior to any user data collection.

8. Risk Assessment

Technical risks include model underperformance and integration bugs, mitigated through iterative testing and benchmarking against the Wav2Vec2 alternative. Data-related risks include limited dataset diversity, mitigated by combining RAVDESS and TESS for broader coverage. Operational risks include time constraints across a four-person team, mitigated through clear task allocation by component (recognition, memory, integration, evaluation) and weekly progress reviews with supervisors.

9. Timeplan



10. Conclusion

This project proposes an emotionally adaptive AI companion that advances beyond existing systems through voice-based emotion recognition, structured longitudinal emotional memory, and explainable personalized profiling. By formulating three testable, non-redundant hypotheses and rigorous evaluation metrics, this research aims to demonstrate measurable improvement over current AI companions such as Replika and ChatGPT.

11. References

- [1] A. Pruss and M. Alimardani, “Emotional Coregulation in Close Relationships with AI Agents: A Survey of ChatGPT and Replika Users,” in Proceedings of the ACM on Human-Computer Interaction, 2025, doi: 10.1145/3765766.3765801.
- [2] A. Manoli, J. V. T. Pauketat, A. Ladak, H. Noh, A. H.-C. Hwang, and J. R. Anthis, “Digital Companionship: Overlapping Uses of AI Companions and AI Assistants,” in Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), New York, NY, USA: Association for Computing Machinery, 2026, Article 372, pp. 1–25, doi: 10.1145/3772318.3791331.